



Gaussian Sum Filters for Space Surveillance: Theory and Simulations

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While standard Kalman-based filters, Gaussian assumptions, and covariance-weighted metrics are very effective in data-rich tracking environments, their use in the data-sparse environment of space surveillance is more limited. To properly characterize non-Gaussian density functions arising in the problem of long-term propagation of state uncertainties, a Gaussian sum filter adapted to the two-body problem in space surveillance is proposed and demonstrated to achieve uncertainty consistency. The proposed filter is made efficient by using only a one-dimensional Gaussian sum in equinoctial orbital elements, thereby avoiding the expensive representation of a full six-dimensional mixture and hence the “curse of dimensionality.” Additionally, an alternate set of equinoctial elements is proposed and is shown to provide enhanced uncertainty consistently over the traditional element set. Simulation studies illustrate the improvements in the Gaussian sum approach over the traditional unscented Kalman filter and the impact of correct uncertainty representation in the problems of data association (correlation) and anomaly (maneuver) detection.

Nomenclature

A	= lower-triangular Cholesky factor of a covariance matrix
a	= semimajor axis
$\mathbf{a}_{\text{pert}}$	= perturbing acceleration
(a, h, k, p, q, ℓ)	= equinoctial orbital elements
c	= parameter controlling the accuracy of the Gaussian sum filter
$\mathbf{f}$	= system dynamics vector
G	= process noise shape matrix
h	= measurement function vector
k	= (subscript) time index
ℓ	= mean longitude
N	= number of mixture components
$\mathcal{N}$	= Gaussian probability density function
n	= mean motion
(n, h, k, p, q, ℓ)	= alternate equinoctial orbital elements
P	= covariance of a Gaussian distribution
$\mathcal{PE}$	= prediction error
p	= probability density function
Q(t)	= process noise covariance matrix
R_k	= measurement noise covariance matrix
r	= Cartesian Earth-Centered Inertial (ECI) position coordinates
$\dot{\mathbf{r}}$	= Cartesian ECI velocity coordinates
$\ddot{\mathbf{r}}$	= Cartesian ECI acceleration coordinates
t_0, t	= initial and current times
u	= orbital element coordinates (six-dimensional)
$\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$	= Gaussian white noise sequence
$\mathbf{w}(t)$	= Gaussian white noise process
$w_\alpha, w_\beta, \dots$	= mixture weights
x	= dynamic state vector
$\mathbf{Z}_k = \{\mathbf{z}_1, \dots, \mathbf{z}_k\}$	= measurement sequence

$\alpha, \beta, \dots$	= (subscripts) indices of mixture components
δ_{kj}	= Kronecker delta symbol
μ	= mean of a Gaussian distribution
$\mu_\oplus$	= Earth gravitational constant (398600.4418 km ³ /s ²)
σ	= standard deviation of a univariate Gaussian distribution
ψ	= inverse solution flow
∇_x	= gradient operator with respect to $\mathbf{x}$ (column oriented)

Introduction

SPACE situational awareness (SSA) encompasses intelligence, reconnaissance of all space objects, and the prediction of space events, possible collisions, threats, and activities. Fundamental to SSA are the problems of data/track association (correlation), conjunction analysis, sensor resource management, and anomaly (e.g., maneuver, change) detection. Common amongst these tracking problems is the requirement of *uncertainty consistency*, which is the accurate and truthful representation of the probability density function (PDF) of each resident space object (RSO) in the space catalog. Within the problems of nonlinear estimation and filtering, uncertainty consistency entails computing an accurate finite-dimensional representation of a (possibly non-Gaussian) PDF and accurately implementing the prediction and fusion steps of the general Bayesian state estimator [1].

The problem of tracking RSOs and correctly managing and representing their uncertainties presents some unique and formidable challenges not found in other tracking environments. In contrast to air or ground tracking, the space surveillance environment is *data-starved*. Typical RSO tracking problems require the long-term propagation of state PDFs, often on the order of several orbital periods, using high-fidelity dynamical models in the absence of measurement or track updates. Often, the state uncertainty of an RSO is assumed to be Gaussian which, although may be appropriate initially, will inevitably become non-Gaussian if propagated for a sufficiently long time span under nonlinear dynamics (i.e., gravity, drag, solar radiation, third-body perturbations, etc.). This departure from “Gaussianity” plagues the standard Kalman-based filtering algorithms such as those of the extended (EKF) and unscented (UKF) variety. More accurate filters which better approximate model nonlinearities and non-Gaussian PDFs are therefore sought. The

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Gaussian sum filter is one such example which has been investigated in SSA applications by the authors [2,3] and other researchers [4–8].

The Gaussian sum filter (GSF) is based on a fundamental result of Alspach and Sorenson [4], which states that any PDF can be approximated arbitrarily close (in the L^1 sense) by a weighted sum (mixture) of Gaussian PDFs henceforth called a Gaussian sum. Thus, Gaussian sums provide a mechanism for modeling non-Gaussian densities and for more accurately approximating the solution of the Fokker–Planck–Kolmogorov equation (FPKE) [9], which governs the time evolution of a PDF under a nonlinear stochastic dynamical system. Computationally, the GSF has the added advantage of being parallelizable since filters such as the EKF or UKF act *independently* on each component Gaussian in the prediction and correction steps. With regard to updating the weights within the filter, such a scheme is clearly dictated from Bayes' rule following a measurement or track update (fusion). However, there is not complete agreement on how the weights should be updated (if at all) following a prediction. Our philosophy is to *not* update the weights, which is justified because uncertainty consistency is achieved by working in coordinates *adapted to the dynamics* (i.e., orbital elements). On the other hand, some researchers [6–8] have proposed methods for adapting the weights based on various L^2 optimization criteria, for example, using the FPKE error as feedback. The impact of using one these weight update schemes during propagation is analyzed in this paper. It is found that the method does not improve uncertainty consistency when applied to the proposed GSF, but rather causes the accuracy of the Gaussian sum representation to degrade slightly.

This paper builds on the earlier works of Horwood and Poore [2] and Horwood et al. [3], which provide an initial framework for a GSF based on efficient refinement and coarsening schemes (i.e., adaptively changing the number of mixture components). Although the previous methodology is very general and applicable to a variety of tracking problems, it suffers from the “curse of dimensionality” because the refinement operation is performed along all dimensions of the distribution. In the six-dimensional setting of space surveillance, the length of the Gaussian sum generated from this algorithm would be prohibitive. In the present paper, we develop a new and more efficient GSF by noting that, in *equinoctial orbital element space*, one can accurately propagate the full state PDF using only a *one-dimensional* Gaussian sum. This performance enhancement is not possible in a system of Cartesian Earth-centered inertial (ECI) coordinates because the nonlinearities of the dynamics act strongly in all six coordinates. In contrast, in orbital element space (equinoctial or otherwise), five of the six coordinates evolve *linearly* under the unperturbed two-body (Kepler) problem and evolve *nearly linearly* under perturbed dynamics. Consequently, the uncertainty grows mainly along one coordinate (the in-track or mean longitude direction). Dependent on the size of the initial uncertainty in the radial direction (semimajor axis) and the propagation time, a one-dimensional Gaussian sum in equinoctial space can be defined. Furthermore, when the UKF is used to propagate each component Gaussian in parallel, the propagated Gaussian sum is argued to be a consistent representation of the true PDF. The underlying GSF is made very efficient because the initial Gaussian sum is computed by solving an L^2 optimization problem once *offline* (the solution is also demonstrated to be near optimal in L^1 and L^∞) and used thereafter in any scenario through a lookup table.

An additional finding communicated in this paper is that enhanced uncertainty consistency can be achieved by representing state PDFs in “alternate” equinoctial orbital elements, which use the mean motion in place of the semimajor axis. Under unperturbed Keplerian dynamics, *all six* of these coordinates transform linearly. To the knowledge of the authors, this alternate set of equinoctial elements does not appear to be recognized within the space surveillance community.

To demonstrate the impact of correct uncertainty characterization of an RSO's state in the problem of long-term propagation, the prediction error component of the likelihood ratio used to score the association of a track to an orbit has been evaluated using a high-fidelity gravity model in conjunction with the new GSF. Because the statistically correct prediction error can be accurately computed, the

GSF is shown to maintain uncertainty consistency over many orbital periods. This is compared with the classical covariance-weighted prediction error which, while sufficient for short-term prediction in a data-rich environment, is demonstrated to be inadequate for long-term prediction required in the data-sparse space surveillance environment. Simulation studies are also performed illustrating the impact of correct uncertainty management on the problems of data association (correlation) and anomaly detection.

The plan of the paper is as follows. First, the coordinate systems used in this paper are defined including the alternate system of equinoctial orbital elements. Second, the mathematical theory of nonlinear filtering is summarized and subsequently specialized to mixture PDFs. Third, the new Gaussian sum filter is developed showing precisely how the initial Gaussian sum is defined so that uncertainty consistency is maintained when it is propagated over a long time gap. Fourth, an online metric based on the prediction error is proposed, which provides a means to assess track state uncertainty consistency. Finally, results are presented and conclusions are made. The appendix summarizes a weight update scheme for uncertainty propagation proposed originally by Terejanu et al. [6], which is subsequently validated against the new GSF in the simulations section.

Coordinate Systems

In this section, the proposed orbital element coordinate systems used in this work are briefly outlined. To begin, note that with respect to Cartesian ECI position-velocity coordinates $(\mathbf{r}, \dot{\mathbf{r}})$, the acceleration $\ddot{\mathbf{r}}$ of a resident space object (satellite, debris) can be written in the form

$$\ddot{\mathbf{r}} = -\frac{\mu_\oplus}{r^3}\mathbf{r} + \mathbf{a}_{\text{pert}}(\mathbf{r}, \dot{\mathbf{r}}, t) \quad (1)$$

In this equation, $r = |\mathbf{r}|$, $\mu_\oplus = GM_\oplus$ where G is the gravitational constant and $M_\oplus$ is the mass of the Earth, and $\mathbf{a}_{\text{pert}}$ encapsulates all perturbing accelerations of the space object other than those due to the two-body point mass gravitational acceleration.

The *equinoctial orbital elements* [10] (a, h, k, p, q, ℓ) define a system of curvilinear coordinates in position-velocity space. Physical and geometric interpretations of these coordinates as well as the transformation from equinoctial elements to ECI are provided, for example, in Montenbruck and Gill [11] and Vallado [12]. Models for the perturbing acceleration $\mathbf{a}_{\text{pert}}$ are also developed in these references. The authors propose the definition of *alternate equinoctial orbital elements* (n, h, k, p, q, ℓ) , in which the mean motion $n = \sqrt{\mu_\oplus/a^3}$ is used as the first coordinate in place of the semimajor axis a .

The representation of the dynamical model (1) in coordinate systems other than ECI is straightforward to obtain using the chain rule. Indeed, if $\mathbf{u} = \mathbf{u}(\mathbf{r}, \dot{\mathbf{r}})$ denotes a coordinate transformation from ECI position-velocity coordinates $(\mathbf{r}, \dot{\mathbf{r}})$ to coordinates $\mathbf{u} \in \mathbb{R}^6$ (e.g., equinoctial or alternate equinoctial), then Eq. (1) is transformed to

$$\dot{\mathbf{u}} = \dot{\mathbf{u}}_{\text{Kepler}} + \frac{\partial \mathbf{u}}{\partial \mathbf{r}} \cdot \mathbf{a}_{\text{pert}}(\mathbf{r}, \dot{\mathbf{r}}, t) \quad (2)$$

where

$$\dot{\mathbf{u}}_{\text{Kepler}} = \frac{\partial \mathbf{u}}{\partial \mathbf{r}} \cdot \dot{\mathbf{r}} - \frac{\mu_\oplus}{r^3} \frac{\partial \mathbf{u}}{\partial \dot{\mathbf{r}}} \cdot \mathbf{r} \quad (3)$$

If $\mathbf{u}$ is either equinoctial or alternate equinoctial orbital elements, then Eq. (3) simplifies to

$$\dot{\mathbf{u}}_{\text{Kepler}} = (0, 0, 0, 0, 0, n)^T \quad (4)$$

where n is the mean motion. Notice that if $\mathbf{a}_{\text{pert}} = \mathbf{0}$, then Eq. (2) is a *linear* system of differential equations in the alternate equinoctial elements but not in the equinoctial elements. In the latter coordinate system, the solution of Eq. (4), though trivial, depends nonlinearly on the semimajor axis since $\ell(t) = \ell_0 + \sqrt{\mu_\oplus/a_0^3}(t - t_0)$. Therefore, a PDF that is initially Gaussian in alternate equinoctial elements will

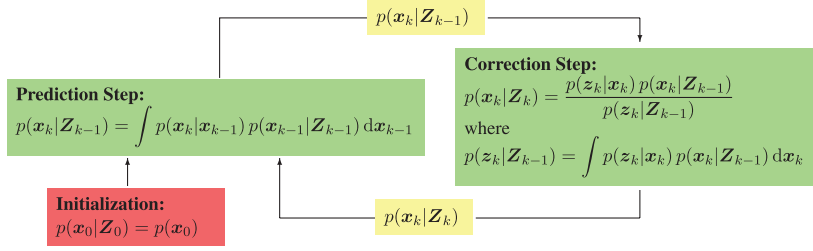


Fig. 1 The predictor-corrector step for the recursive Bayesian state estimator.

remain identically Gaussian when propagated under unperturbed Keplerian dynamics. This same conclusion does not hold in the (traditional) equinoctial orbital elements.

Overview of Nonlinear Filtering

In this section, the general Bayesian framework for nonlinear filtering is reviewed and the filter prediction step is subsequently specialized to general mixture probability density functions (PDFs). Further specializations to Gaussian mixtures are relegated to the next section.

The problem of nonlinear filtering requires the definition of dynamical and measurement models. It is assumed that the dynamic state $\mathbf{x}(t) \in \mathbb{R}^n$ at time t evolves according to the continuous-time stochastic model

$$\dot{\mathbf{x}}'(t) = \mathbf{f}(\mathbf{x}(t), t) + \mathbf{G}(\mathbf{x}(t), t)\mathbf{w}(t) \quad (5)$$

where $\mathbf{f}: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$, $\mathbf{G}: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^{n \times m}$, and $\mathbf{w}(t)$ is an m -dimensional Gaussian white noise process with covariance matrix $\mathbf{Q}(t)$. In particular in Eq. (5), the function $\mathbf{f}$ encodes the deterministic force components of the dynamics (e.g., gravity, drag, etc.) while the process noise term models the stochastic acceleration. In many tracking applications, it is often convenient to work with a discrete-time formulation of the dynamical model which assumes the form

$$\mathbf{x}_{k+1} = \mathbf{f}_k(\mathbf{x}_k) + \mathbf{G}_k(\mathbf{x}_k)\mathbf{w}_k \quad (6)$$

where $\mathbf{x}_k = \mathbf{x}(t_k)$, $\mathbf{f}_k: \mathbb{R}^n \rightarrow \mathbb{R}^n$, $\mathbf{G}_k: \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$, and $\{\mathbf{w}_k\}$ is an m -dimensional zero-mean Gaussian white noise sequence with covariance matrix $\mathbf{Q}_k$. In space surveillance, the process noise term is often very small and discarded. In such situations, the function $\mathbf{f}_k$ is just the solution flow corresponding to the continuous model (5) with $\mathbf{w}(t) = 0$.

A sequence of measurements $\mathbf{Z}_k \equiv \{\mathbf{z}_1, \dots, \mathbf{z}_k\}$ is related to the corresponding kinematic states $\mathbf{x}_k$ via measurement functions $\mathbf{h}_k: \mathbb{R}^n \rightarrow \mathbb{R}^p$ according to the discrete-time measurement model

$$\mathbf{z}_k = \mathbf{h}_k(\mathbf{x}_k) + \mathbf{v}_k \quad (7)$$

In this equation, $\{\mathbf{v}_k\}$ is a p -dimensional zero-mean Gaussian white noise sequence with covariance matrix $\mathbf{R}_k$. More general filter models can be formulated from measurement models with non-Gaussian or correlated (e.g., colored) noise terms [13] and sensor biases [14].

In the Bayesian approach to dynamic state estimation, one constructs the *posterior* PDF of the state based on information of a *prior* state and received measurements. Encapsulating all available statistical information, the posterior PDF $p(\mathbf{x}_k|\mathbf{Z}_k)$ may be regarded as the complete solution to the estimation problem and various optimal state estimates can be computed from it. The prediction and correction filter steps are summarized in the flowchart of Fig. 1 (with additional details provided, for example, in Ristic et al. [1]).

Analytical solutions to the filter prediction and correction steps in Fig. 1 are generally intractable and are only known in a few restrictive cases. In practice, models are nonlinear and states can be non-Gaussian; one must be content with an approximate or suboptimal algorithm for the Bayesian state estimator. While the EKF and UKF are used extensively in air and missile tracking, they only represent state uncertainties by a covariance matrix and this may not be

adequate in the space surveillance environment. Because of the need to propagate uncertainties over extended time intervals in the absence of measurement updates, higher-order cumulants (e.g., skewness, excess kurtosis) can become nonnegligible and must be accounted for to achieve uncertainty consistency.

Specialization to Mixture Densities

Given a set of PDFs $p_1(\mathbf{x}), \dots, p_N(\mathbf{x})$, called the mixture components, and weights $w_1, \dots, w_N$ such that $w_\alpha \geq 0$, for $\alpha = 1, \dots, N$, and $\sum_{\alpha=1}^N w_\alpha = 1$, the sum

$$p(\mathbf{x}) = \sum_{\alpha=1}^N w_\alpha p_\alpha(\mathbf{x}) \quad (8)$$

is called a *mixture density*. In practice, the mixture components are members of a parametric family of distributions (such as Gaussians) each allowed to have different parameter values. To derive the filter prediction step for a mixture density,[§] suppose the prior density $p(\mathbf{x}_{k-1}|\mathbf{Z}_{k-1})$ at time t_{k-1} is a mixture of the form

$$p(\mathbf{x}_{k-1}|\mathbf{Z}_{k-1}) = \sum_{\alpha=1}^N w_\alpha p_\alpha(\mathbf{x}_{k-1}|\mathbf{Z}_{k-1}) \quad (9)$$

Applying the Chapman–Kolmogorov equation (i.e., the prediction step equation in Fig. 1) independently on each mixture component yields

$$p_\alpha(\mathbf{x}_k|\mathbf{Z}_{k-1}) = \int p(\mathbf{x}_k|\mathbf{x}_{k-1}) p_\alpha(\mathbf{x}_{k-1}|\mathbf{Z}_{k-1}) d\mathbf{x}_{k-1} \quad (10)$$

The predicted mixture PDF is found to be

$$\begin{aligned} p(\mathbf{x}_k|\mathbf{Z}_{k-1}) &= \int p(\mathbf{x}_k|\mathbf{x}_{k-1}) p(\mathbf{x}_{k-1}|\mathbf{Z}_{k-1}) d\mathbf{x}_{k-1} \\ &= \int p(\mathbf{x}_k|\mathbf{x}_{k-1}) \sum_{\alpha=1}^N w_\alpha p_\alpha(\mathbf{x}_{k-1}|\mathbf{Z}_{k-1}) d\mathbf{x}_{k-1} \\ &= \sum_{\alpha=1}^N w_\alpha p_\alpha(\mathbf{x}_k|\mathbf{Z}_{k-1}) \end{aligned} \quad (11)$$

as follows from Eqs. (9) and (10).

The above analysis shows that the mixture weights remain invariant under uncertainty propagation by a simple consequence of linearity.[¶] However, there is no universal agreement on how the mixture weights should be updated (if at all) following a prediction. For the Gaussian sum propagation algorithm developed in the next section, the prior Gaussian components $p_\alpha(\mathbf{x}_{k-1}|\mathbf{Z}_{k-1})$ are represented in equinoctial orbital elements and “seeded” in such a way that the UKF prediction step accurately approximates the solution of Eq. (10). As shown in the simulations section, this methodology provides excellent uncertainty consistency without the added

[§]The new work presented in this manuscript deals primarily with the prediction step of the Gaussian sum filter. Details on the correction step can be found, for example, in [2,5].

[¶]An analogous conclusion on the invariance of the mixture weights applies if assuming continuous-time dynamics and time evolution governed by the Fokker–Planck–Kolmogorov equation.

expense of updating the weights. Notwithstanding these comments, it is of interest to study if there is any *added benefit* by applying a weight adaptation procedure to the proposed Gaussian sum filter. Details of the specific weight update scheme evaluated in this paper (which is taken from Terejanu et al. [6]) are summarized in the appendix and the results of its application are presented in the simulations section.

Gaussian Sum Propagation

In the previous section, it was shown how the Bayesian nonlinear filter specializes to general mixture densities. In this section, the methodology is further specialized to Gaussian mixtures (Gaussian sums) adapted to the two-body problem.

It is convenient to define the function $\mathcal{N}: \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$ given by

$$\mathcal{N}(\mathbf{x}; \mathbf{y}, \mathbf{P}) = \frac{1}{\sqrt{\det(2\pi\mathbf{P})}} \exp\left[-\frac{1}{2}(\mathbf{x} - \mathbf{y})^T \mathbf{P}^{-1}(\mathbf{x} - \mathbf{y})\right] \quad (12)$$

Note that if the second and third arguments of $\mathcal{N}$ are fixed and the latter is symmetric and positive definite, then $\mathcal{N}$ defines a multivariate Gaussian PDF with respect to its first argument. A Gaussian sum is a mixture density of the form (8) given by

$$p(\mathbf{x}) = \sum_{\alpha=1}^N w_{\alpha} \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_{\alpha}, \mathbf{P}_{\alpha})$$

where $\boldsymbol{\mu}_{\alpha}$ and $\mathbf{P}_{\alpha}$, $\alpha = 1, \dots, N$, are the respective means and covariances of the component Gaussians.

Kalman-based filters such as the EKF or UKF are the workhorse of a GSF since they act in parallel on each component Gaussian during the prediction and correction filtering steps. For optimal efficiency and numerical stability, the use of the discrete-time square-root version of the UKF [15] is proposed that does not directly form any covariance matrix and instead uses the (lower-triangular) Cholesky factor (i.e., square root) of the covariance. Higher-order versions of the UKF (also called Gauss–Hermite filters) that make use of efficient Gauss–Hermite quadrature rules due to Genz and Keister [16] are also proposed in [2]. Further, in this paper it is assumed that the initial prior density is Gaussian. This prior density is usually obtained through a batch least-squares (differential correction) procedure which produces a state estimate (mean) and a Gaussian covariance from a sequence of sensor measurement data. For low-Earth-orbit (LEO) radar data (e.g., a 2 min track generated from radar measurements every ten seconds), it is observed that the initial PDF is well approximated by a single Gaussian in equinoctial orbital elements [17] or ECI coordinates [18]. The Gaussian assumption on the prior can be relaxed; a method for determining a Gaussian sum representation of the prior density from radar measurements is developed in Horwood and Poore [2].

For the remainder of this section, a new algorithm is proposed that efficiently refines a single Gaussian in equinoctial orbital elements into a Gaussian sum. The component means, covariances, and weights of the mixture are chosen such that each Gaussian component remains Gaussian (up to a prescribed accuracy) when propagated by the (square-root version of the) UKF under the nonlinear dynamics. The refinement methodology is illustrated in Fig. 2. Under a nonlinear transformation, a Gaussian (represented by the thick black ellipse) need not be mapped to a Gaussian (e.g., the

level surfaces of the transformed distribution could look crescent-shaped). However, in a sufficiently small neighborhood, any (smooth) nonlinear map will be approximately linear. Consequently, Gaussians with smaller covariances (represented by the colored elliptic disks) remain more Gaussian than those with larger covariances under the nonlinear mapping. Therefore, a Gaussian sum refined by approximating each constituent Gaussian by a finer Gaussian sum will exhibit better behavior through nonlinear transformations.

The propagation of uncertainty under the unperturbed two-body problem [i.e., Keplerian dynamics with $\mathbf{a}_{\text{pert}} = \mathbf{0}$ in Eq. (1)] provides the motivation for the new Gaussian sum refinement method. The time evolution of the equinoctial orbital elements (a, h, k, p, q, ℓ) under the assumed (unperturbed) dynamics is

$$\begin{aligned} a(t) &= a_0, & h(t) &= h_0, & k(t) &= k_0, & p(t) &= p_0 \\ q(t) &= q_0, & \ell(t) &= \ell_0 + n_0(t - t_0) \end{aligned} \quad (13)$$

where $n_0 = \sqrt{\mu_{\oplus}/a_0^3}$ is the mean motion at the initial epoch. Suppose an initial Gaussian at time t_0 has mean $\boldsymbol{\mu}_0$ and covariance $\mathbf{P}_0$. The first key observation is that at any future time $t > t_0$, the covariance between any two of the first five elements remains constant; that is $P^{ij}(t) = P_0^{ij}$ for all $t > t_0$ and $i, j \in \{1, \dots, 5\}$. To leading order, the other components of the propagated covariance matrix are

$$\begin{aligned} P^{i6}(t) &= P_0^{i6} - \frac{3}{2} \frac{n_0}{a_0} P_0^{i1} \Delta t, & i &\in \{1, \dots, 5\} \\ P^{66}(t) &= P_0^{66} - \frac{3}{2} \frac{n_0}{a_0} P_0^{16} \Delta t + \left(\frac{3}{2} \frac{n_0}{a_0} \Delta t\right)^2 P_0^{11} \end{aligned}$$

where $\Delta t = t - t_0$. Note that the covariance between the mean longitude (the sixth equinoctial element) and the other five elements grows *linearly* in time while the variance in the mean longitude grows *quadratically*. Therefore, the variance in the mean longitude $P^{66}(t)$ is the dominant component in the propagated covariance for large Δt . Any departure from Gaussianity would be manifested most strongly in this coordinate. With the inclusion of higher-order perturbative forces in the dynamics, a similar conclusion is conjectured.

The key idea of the new refinement method is to mitigate the growth of the dominant covariance component $P^{66}(t)$ through efficient refinement of the initial Gaussian density. The proposed refinement criterion is now described. For t sufficiently large

$$P^{66}(t) \approx P_0^{66} + \frac{9}{4} \frac{n_0^2}{a_0^2} P_0^{11} (\Delta t)^2 \quad (14)$$

In Eq. (14), the only component of the initial covariance which drives the growth of $P^{66}(t)$ is P_0^{11} , the initial variance in the first equinoctial element, namely the semimajor axis. In other words, *it is the uncertainty along the radial (semimajor axis) direction which causes the uncertainty along the in-track (mean longitude) direction to grow*. Therefore, it is proposed to refine the initial Gaussian at time t_0 only along the semimajor axis a coordinate. This proposed refinement scheme also mitigates the growth of the cross-terms $P^{i6}(t)$.

To mitigate the growth of the uncertainty along the dominant in-track (mean longitude) direction, the initial Gaussian $\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_0, \mathbf{P}_0)$ is refined along only the first dimension (semimajor axis) such that each mixture component has a variance in a of $\tilde{P}_0^{11} < P_0^{11}$. By Eq. (14), the variance in the mean longitude of each component at time t , $\tilde{P}^{66}(t)$, is approximately

$$\tilde{P}^{66}(t) \approx P_0^{66} + \frac{9}{4} \frac{\tilde{n}_0^2}{\tilde{a}_0^2} \tilde{P}_0^{11} (\Delta t)^2 \quad (15)$$

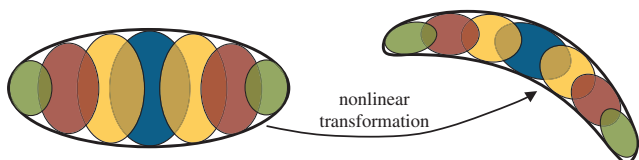


Fig. 2 Depiction of a single Gaussian and its Gaussian sum approximation undergoing a nonlinear transformation.

The new variance $\tilde{P}_0^{11}$ is chosen such that

$$\tilde{P}^{66}(t) \leq c P_0^{66} \quad (16)$$

for some constant $c > 1$. From Eq. (15), it follows that

$$\tilde{P}_0^{11} = \frac{4(c-1)\tilde{a}_0^2}{9\tilde{n}_0^2(\Delta t)^2} P_0^{66} \quad (17)$$

The parameter c in the criterion Eq. (16) is a tuning parameter of the filter that controls the growth of the component covariances and ultimately the accuracy of the underlying Gaussian sum. For example, if $c = 2$, then Eq. (16) in conjunction with Eq. (17) says that the component Gaussians are chosen so that the variance in the mean longitude ℓ never exceeds twice the original variance in ℓ over the propagation time Δt . It is noted from Eq. (17) that for a fixed choice of c , the required variance in the semimajor axis of each component is proportional to the inverse square of the propagation time; longer propagations therefore require finer Gaussian sums.

What remains to be shown in this section is how to refine a single Gaussian into a Gaussian sum where each component has a prescribed mean and covariance. Precisely how this is done is a delicate matter. The theory of Gaussian sums is based on the L^1 norm [4]; however, any optimization criterion based on this norm would likely be computationally impractical due to the need to solve a nonsmooth nonlinear optimization problem. A suboptimal solution based on moment matching is proposed in [2], but the method tends to place too many of the component Gaussians far out on the tails, thereby sacrificing resolution around the center. In what follows, an alternative algorithm for Gaussian sum refinement is proposed by working in L^2 because an analytic expression for the L^2 norm of a Gaussian sum is well known. In what follows, it is first shown how to refine the one-dimensional unit Gaussian $\mathcal{N}(x; 0, 1)$. The refinement method is then extended to the multivariate case with an arbitrary covariance. Although L^2 is the assumed norm, it is shown that the new algorithm exhibits nice convergence properties not only in L^2 but also in L^1 and L^∞ . The general refinement method is summarized as Algorithm 1 at the end of this section.

Refinement of the One-Dimensional Unit Gaussian

A Gaussian sum approximation of the unit one-dimensional Gaussian $\mathcal{N}(x; 0, 1)$ is now derived. A key feature of the ensuing algorithm is that it can be applied *offline* and subsequently used online to refine any univariate or multivariate Gaussian with an arbitrary covariance. The problem has an obvious trivial solution which perfectly approximates the target, namely the Gaussian sum with a single component. However, as motivated earlier, the idea is to construct a Gaussian sum approximation whose component standard deviations σ_α are smaller than some fixed value $\sigma < 1$ so that the Gaussian sum more accurately represents the true distribution under a nonlinear transformation. This requirement leads to a constrained nonlinear optimization problem. That said, it is assumed that the target Gaussian sum has the form

$$p_{\text{GS}}(x) = \sum_{\alpha=1}^N w_\alpha \mathcal{N}(x; \mu_\alpha, \sigma_\alpha^2)$$

for some predetermined length $N > 1$. Now define the objective function

$$E = \frac{1}{2} \|\mathcal{N}(x; 0, 1) - p_{\text{GS}}(x)\|_{L^2}^2 = \frac{1}{2} \int_{-\infty}^{\infty} \left[\mathcal{N}(x; 0, 1) - \sum_{\alpha=1}^N w_\alpha \mathcal{N}(x; \mu_\alpha, \sigma_\alpha^2) \right]^2 dx \quad (18)$$

The weights w_α , means μ_α , and standard deviations σ_α of each component are determined from the solution of the following L^2 optimization problem:

$$\text{Minimize}_{\substack{w_1, \dots, w_N \\ \mu_1, \dots, \mu_N \\ \sigma_1, \dots, \sigma_N}} E$$

$$\text{Subject to } \sum_{\alpha=1}^N w_\alpha = 1, \quad w_\alpha \geq 0, \quad \alpha = 1, \dots, N$$

$$\mu_1 \leq \mu_2 \leq \dots \leq \mu_N$$

$$\sigma_\alpha \leq \sigma < 1, \quad \alpha = 1, \dots, N \quad (19)$$

Noting the identity

$$\int \mathcal{N}(x; \mu_1, \mathbf{P}_1) \mathcal{N}(x; \mu_2, \mathbf{P}_2) dx = \mathcal{N}(\mu_1 - \mu_2; \mathbf{0}, \mathbf{P}_1 + \mathbf{P}_2) \quad (20)$$

it follows that Eq. (18) reduces to

$$E = \frac{1}{2} \mathbf{w}^T \mathbf{M} \mathbf{w} - \mathbf{w}^T \mathbf{n} + \frac{1}{4\sqrt{\pi}} \quad (21)$$

where

$$(\mathbf{w})_\alpha = w_\alpha, \quad (\mathbf{n})_\alpha = \mathcal{N}(\mu_\alpha; 0, \sigma_\alpha^2 + 1)$$

$$(\mathbf{M})_{\alpha\beta} = \mathcal{N}(\mu_\alpha - \mu_\beta; 0, \sigma_\alpha^2 + \sigma_\beta^2)$$

the subscripts above denote vector and matrix components.

The minimization of Eq. (21) over the individual weights, means, and standard deviations subject to the constraints in Eq. (19) is a difficult nonlinear optimization problem. (It is not a quadratic programming problem since $\mathbf{M}$ depends on the unknown means and standard deviations.) Further insight into this general problem is provided at the end of this subsection. In what follows, a suboptimal yet computationally tractable algorithm is proposed which approximates the solution of Eq. (19). This procedure is summarized in Algorithm 1 at the end of the section. Specifically, each of the component Gaussians is constrained to have a *common standard deviation* $\sigma_\alpha = \sigma < 1$ and the *means* μ_α are fixed according to

$$\mu_\alpha = -m + \sigma(\alpha - 1) \quad (22)$$

for $\alpha = 1, \dots, N$, where $m > 0$ and $N = \lceil 1 + 2m/\sigma \rceil$. Note that the means are evenly distributed with the left-most and right-most components located at $x \approx \pm m$. To ensure adequate accuracy around the tails, it is suggested to set $m = 4$ if $\sigma \geq \frac{1}{2}$, otherwise $m = 6$. With these additional constraints, the optimization problem (19) reduces to a quadratic programming problem which is straightforward to solve using elementary methods.

Figure 3 shows plots of the L^1 , L^2 , and L^∞ error between the unit one-dimensional Gaussian and its Gaussian sum approximation computed using the suboptimal approach described above. Up to around $N \approx 100$, the three L^p errors all decrease, but they plateau to

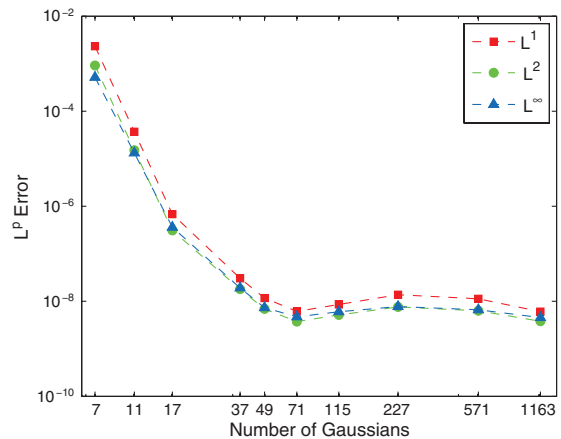


Fig. 3 Plots of the L^p error between the unit one-dimensional Gaussian and its Gaussian sum approximation obtained using Algorithm 1.

around 10^{-8} for $N \gtrsim 100$. This plateau is believed not to arise because of numerical stability issues [the matrix $\mathbf{M}$ in Eq. (21) has a condition number on the order of 10^5] but simply because *the standard deviations and means are fixed* via Eq. (22) and the minimization is performed only over the unknown weights; this need not be optimal. Notwithstanding these comments, the simulation studies considered in this paper use this refinement scheme with N as high as 1000 and no adverse effects have been observed. If one insists on computing a Gaussian sum approximation of $\mathcal{N}(x; 0, 1)$ with an L^p error much less than 10^{-8} , one could replace the component mean locations (22) by

$$\mu_\alpha = -m + \frac{1}{2}\sigma(\alpha - 1)$$

for $\alpha = 1, \dots, N$, where $N = \lceil 1 + 4m/\sigma \rceil$. If, for example, one takes $\sigma = \frac{1}{5}$ and $m = 6$, then $N = 121$ and the L^1 , L^2 , and L^∞ errors are $1.8 \cdot 10^{-11}$, $7.7 \cdot 10^{-12}$, and $8.2 \cdot 10^{-12}$, respectively. If $\sigma = \frac{1}{10}$ and $m = 8$, then $N = 321$ and the corresponding L^p errors diminish to $4.7 \cdot 10^{-18}$, $2.9 \cdot 10^{-18}$, and $4.5 \cdot 10^{-18}$. In both examples, the corresponding condition number of the matrix $\mathbf{M}$ in Eq. (21) is on the order of 10^{16} ; thus quadruple precision or software floating-point arithmetic is required.

To conclude this subsection, some preliminary solutions of the full optimization problem (19) are presented. It is observed that the numerical conditioning of this problem worsens as the target length N of the Gaussian sum increases thereby rendering double precision floating-point arithmetic largely inadequate. To facilitate an accurate solution, the MAPLE Optimization package has been used which takes advantage of built-in library routines provided by the Numerical Algorithms Group with the ability to call them within an arbitrary-precision software floating-point environment.

Figure 4 shows various scatter plots of the component weights and means computed using two different methods. The first method uses the suboptimal approach which fixes the means according to Eq. (22)

and subsequently optimizes over the weights only. The second method solves the full optimization problem (19). Each subplot assumes a different target length N and standard deviation σ . One interesting observation is that the optimal component standard deviations lie on the constraint boundary; i.e., $\sigma_\alpha = \sigma$ for all α . Most importantly, by solving the full optimization problem, the locations of the means are no longer uniformly distributed and the resulting L^2 error is smaller than that obtained by solving the suboptimal problem. In particular, to get the L^2 error down to 10^{-7} requires about $N = 28$ Gaussians using the suboptimal method but only about $N = 9$ Gaussians when solving the full optimization problem. Thus, with the optimal solution of Eq. (19), the error in the Gaussian sum approximation can be substantially reduced or, for the same error, the number of Gaussians needed to achieve a prescribed accuracy can be reduced. It must be emphasized that the full problem, although very expensive to solve, can nevertheless be done *offline*. One can generate a library of Gaussian sum approximations to $\mathcal{N}(x; 0, 1)$ for various values of N and σ . Such computations are in progress. Future work will also attempt to solve the analog of Eq. (19) in L^1 , the norm of which the theory of Gaussian sums is based [4].

Refinement of a Multivariate Gaussian

Using the scheme of the previous subsection, assume a Gaussian sum approximation of the unit one-dimensional Gaussian of the form

$$\mathcal{N}(x; 0, 1) \approx \sum_{\alpha=1}^N w_\alpha \mathcal{N}(x; \mu_\alpha, \sigma^2) \quad (23)$$

where $\sigma^2 < 1$. By linearity, Eq. (23) induces a Gaussian sum approximation of the Gaussian with mean v and variance ρ^2 according to

$$\mathcal{N}(x; v, \rho^2) \approx \sum_{\alpha=1}^N w_\alpha \mathcal{N}(x; v + \rho\mu_\alpha, \sigma^2\rho^2) \quad (24)$$

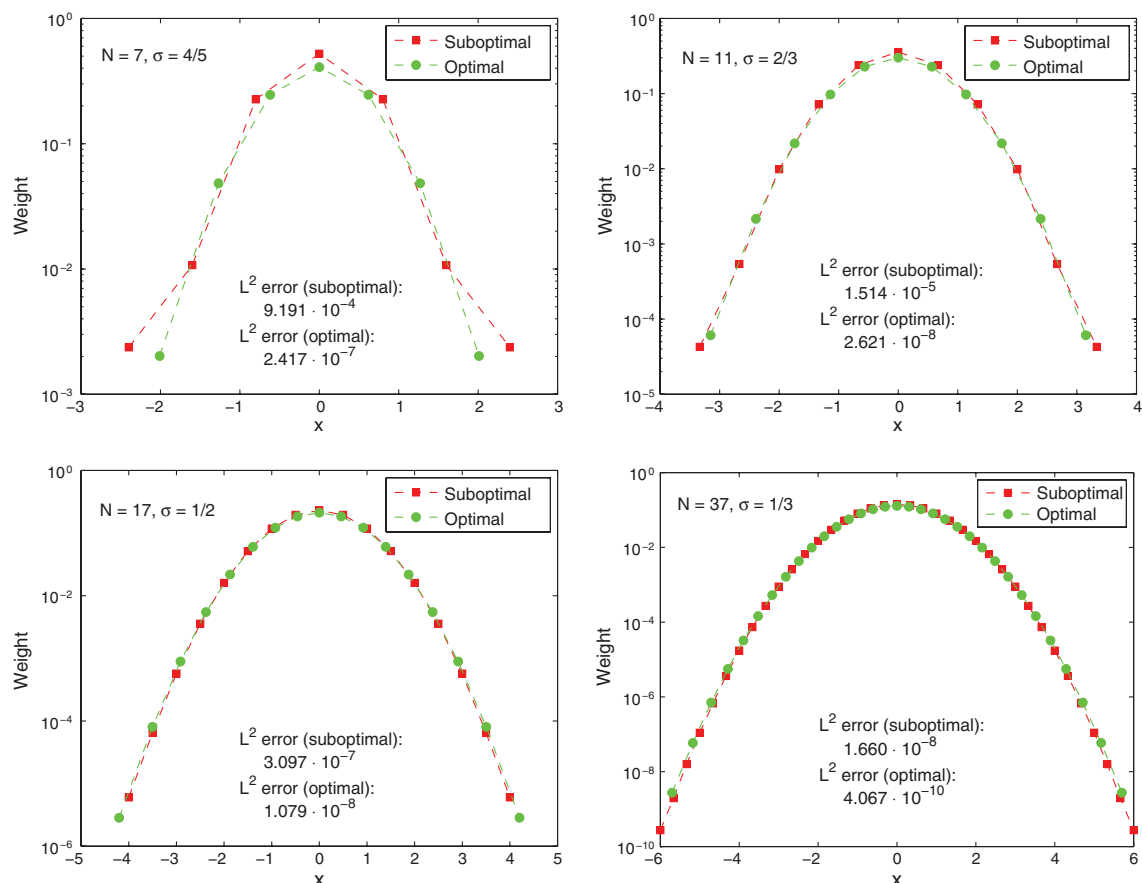


Fig. 4 Scatter plots of the component weights and means computed using the suboptimal algorithm (i.e., minimizing only over the weights) and by solving the full (optimal) optimization problem (i.e., minimizing over the weights, means, and standard deviations).

To extend Eq. (24) to the multivariate Gaussian $\mathcal{N}(\mathbf{x}; \mathbf{v}, \mathbf{Q})$, assume without loss of generality that refinement is done along the first coordinate x^1 of $\mathbf{x}$ (this is the semimajor axis if $\mathbf{x}$ denotes an equinoctial orbital element state). To proceed, consider a Gaussian sum approximation of the quotient

$$\frac{\mathcal{N}(x; 0, 1)}{\mathcal{N}(x; 0, 1)} \approx \sum_{\alpha=1}^N w_{\alpha} \frac{\mathcal{N}(x; \mu_{\alpha}, \sigma^2)}{\mathcal{N}(x; 0, 1)} \equiv \sum_{\alpha=1}^N \tilde{w}_{\alpha} \mathcal{N}(x; \tilde{\mu}_{\alpha}; \tilde{\sigma}^2) \quad (25)$$

where $\sigma^2 < 1$ and the weights w_{α} and means μ_{α} are the solution of Eq. (19), as detailed in the previous subsection. It follows that

$$\begin{aligned} \tilde{w}_{\alpha} &= \frac{\sqrt{2\pi}}{\sqrt{1-\sigma^2}} w_{\alpha} \exp\left[-\frac{\mu_{\alpha}^2}{2(1-\sigma^2)}\right] \\ \tilde{\mu}_{\alpha} &= \frac{\mu_{\alpha}}{1-\sigma^2}, \quad \tilde{\sigma}^2 = \frac{\sigma^2}{1-\sigma^2} \end{aligned} \quad (26)$$

Up to a multiplicative constant, the quotient (25) defines a Gaussian sum approximation of the uniform distribution on the interval $(-m, m)$. Therefore, by the linear transformation $x \rightarrow v^1 + \sqrt{Q^{11}}x$, the quotient

$$\frac{\mathcal{N}(x^1; v^1, Q^{11})}{\mathcal{N}(x^1; v^1, Q^{11})} \approx \sum_{\alpha=1}^N \tilde{w}_{\alpha} \mathcal{N}(x^1; \hat{\mu}_{\alpha}, \hat{\sigma}^2) \quad (27)$$

defines (up to a multiplicative constant) a Gaussian sum approximation of the uniform distribution on the interval $(v^1 - m\sqrt{Q^{11}}, v^1 + m\sqrt{Q^{11}})$, where

$$\hat{w}_{\alpha} = \tilde{w}_{\alpha}, \quad \hat{\mu}_{\alpha} = v^1 + \sqrt{Q^{11}}\tilde{\mu}_{\alpha}, \quad \hat{\sigma}^2 = \tilde{\sigma}^2 Q^{11}$$

Thus, to obtain a Gaussian sum approximation of $\mathcal{N}(\mathbf{x}; \mathbf{v}, \mathbf{Q})$ where refinement is done along the first coordinate, the Gaussian PDF $\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \mathbf{Q})$ is multiplied by the Gaussian sum (27) approximating the uniform PDF. This leads to

$$\begin{aligned} \mathcal{N}(\mathbf{x}; \mathbf{v}, \mathbf{Q}) &\approx \sum_{\alpha=1}^N \hat{w}_{\alpha} \mathcal{N}(\mathbf{x}; \mathbf{v}, \mathbf{Q}) \mathcal{N}(x^1; \hat{\mu}_{\alpha}, \hat{\sigma}^2) \\ &\equiv \sum_{\alpha=1}^N \tilde{w}_{\alpha} \mathcal{N}(\mathbf{x}; \bar{\mathbf{v}}_{\alpha}, \bar{\mathbf{Q}}) \end{aligned} \quad (28)$$

where

$$\begin{aligned} \bar{\mathbf{Q}} &= (\hat{\sigma}^{-2} \mathbf{e}_1 \mathbf{e}_1^T + \mathbf{Q}^{-1})^{-1}, \quad \bar{\mathbf{v}}_{\alpha} = \bar{\mathbf{Q}}(\hat{\sigma}^{-2} \hat{\mu}_{\alpha} \mathbf{e}_1 + \mathbf{Q}^{-1} \mathbf{v}) \\ \bar{w}_{\alpha} &= \hat{w}_{\alpha} \mathcal{N}(\hat{\mu}_{\alpha} - \mathbf{e}_1^T \mathbf{v}; 0, \hat{\sigma}^2 + \mathbf{e}_1^T \mathbf{Q} \mathbf{e}_1) \end{aligned} \quad (29)$$

and $\mathbf{e}_1 = (1, 0, \dots, 0)^T \in \mathbb{R}^6$. The final step of the calculation is the renormalization of the weights $\tilde{w}_{\alpha}$ in Eq. (28) via $\tilde{w}_{\alpha} \rightarrow \bar{w}_{\alpha} / \sum_{\beta=1}^N \bar{w}_{\beta}$. The refinement algorithm is summarized below.

Algorithm 1. Refinement of a Gaussian into a Gaussian Sum
Given a Gaussian $\mathcal{N}(\mathbf{x}; \mathbf{v}, \mathbf{Q})$, where $\mathbf{x}$ denotes an equinoctial orbital element state, ** a Gaussian sum approximation

$$\mathcal{N}(\mathbf{x}; \mathbf{v}, \mathbf{Q}) \approx \sum_{\alpha=1}^N \bar{w}_{\alpha} \mathcal{N}(\mathbf{x}; \bar{\mathbf{v}}_{\alpha}, \bar{\mathbf{Q}})$$

is computed as follows.

1) Select the refinement parameter $\sigma \in (0, 1)$ either manually or according to

$$\sigma = \frac{2\sqrt{c-1}v^1}{3nt} \sqrt{\frac{Q^{66}}{Q^{11}}}$$

**Algorithm 1 is also applicable if $\mathbf{x}$ is an alternate equinoctial orbital element state. In such a case, the equation in Step 1 becomes $\sigma = \sqrt{c-1} \sqrt{Q^{66}/Q^{11}}/t$.

where $n = \sqrt{\mu_{\oplus}/(v^1)^3}$, t is an upper bound for the total propagation time, and $c > 1$ is a growth control parameter as explained following Eq. (17).

2) Compute the number of terms N in the Gaussian sum according to $N = \lceil 1 + 2m/\sigma \rceil$, where $m = 4$ if $\sigma \geq \frac{1}{2}$ or $m = 6$ if $\sigma < \frac{1}{2}$.

3) For $\alpha = 1, \dots, N$, compute $\mu_{\alpha} = -m + \sigma(\alpha - 1)$.

4) Compute the matrix $\mathbf{M}$ and vector $\mathbf{n}$ with components

$$(\mathbf{M})_{\alpha\beta} = \mathcal{N}(v_{\alpha} - v_{\beta}; 0, 2\sigma^2), \quad (\mathbf{n})_{\alpha} = \mathcal{N}(v_{\alpha}; 0, \sigma^2 + 1)$$

5) Minimize the quadratic function^{††} $\frac{1}{2} \mathbf{w}^T \mathbf{M} \mathbf{w} - \mathbf{w}^T \mathbf{n}$ over $\mathbf{w}$ subject to the constraints $\sum_{\alpha=1}^N w_{\alpha} = 1$ and $w_{\alpha} \geq 0$, for $\alpha = 1, \dots, N$.

6) For $\alpha = 1, \dots, N$, compute $\tilde{w}_{\alpha}$, $\tilde{\mu}_{\alpha}$, and $\tilde{\sigma}^2$ from Eq. (26) using the means μ_{α} from step 3 and the weights w_{α} from Step 5.

7) For $\alpha = 1, \dots, N$, compute $\hat{\mu}_{\alpha} = v^1 + \sqrt{Q^{11}}\tilde{\mu}_{\alpha}$ and set $\hat{w}_{\alpha} = \tilde{w}_{\alpha}$ and $\hat{\sigma}^2 = \tilde{\sigma}^2 Q^{11}$.

8) Using the data from the previous step, compute $\bar{\mathbf{Q}}$, $\bar{\mathbf{v}}_{\alpha}$, and $\bar{w}_{\alpha}$ from Eq. (29).

9) For $\alpha = 1, \dots, N$, compute the renormalized weight $\bar{w}_{\alpha} \rightarrow \bar{w}_{\alpha} / \sum_{\beta=1}^N \bar{w}_{\beta}$.

Metric for Uncertainty Consistency

The metric derived in this section is motivated from the prediction error term arising in the correction step of the Bayesian state estimator. This term also appears in the likelihood ratios for scoring the association of one report (e.g., track, measurement) to another [19]. Its accurate evaluation is thus critical in the problem of data/track association (correlation). It will be shown and later demonstrated that the prediction error provides an offline metric to validate the accuracy of the filter prediction step and assess uncertainty consistency when “truth” is not available.

Referring to Fig. 1, for *measurement-to-track association*, the prediction error is precisely the denominator appearing in the correction (fusion) step:

$$\mathcal{PE} = p(z_k | \mathbf{Z}_{k-1}) = \int p(z_k | \mathbf{x}_k) p(\mathbf{x}_k | \mathbf{Z}_{k-1}) d\mathbf{x}_k \quad (30)$$

For *track-to-track association*, the predicted (prior) PDF $p(\mathbf{x}_k | \mathbf{Z}_{k-1})$ is updated with a new track $\mathbf{x}_k^{(\text{new})}$ (which is a random variable) to form the corrected (fused, posterior) PDF $p(\mathbf{x}_k | \mathbf{x}_k^{(\text{new})}, \mathbf{Z}_{k-1})$ using Bayes’ rule:

$$p(\mathbf{x}_k | \mathbf{x}_k^{(\text{new})}, \mathbf{Z}_{k-1}) = \frac{p(\mathbf{x}_k^{(\text{new})} | \mathbf{x}_k, \mathbf{Z}_{k-1}) p(\mathbf{x}_k | \mathbf{Z}_{k-1})}{p(\mathbf{x}_k^{(\text{new})} | \mathbf{Z}_{k-1})} \quad (31)$$

Note that Eq. (31) is precisely the correction step in Fig. 1 with the identifications $z_k \leftrightarrow \mathbf{x}_k^{(\text{new})}$ and $\mathbf{Z}_k \leftrightarrow (\mathbf{x}_k^{(\text{new})}, \mathbf{Z}_{k-1})$. Therefore, the corresponding prediction error is the denominator in Eq. (31):

$$\mathcal{PE} = p(\mathbf{x}_k^{(\text{new})} | \mathbf{Z}_{k-1}) = \int p(\mathbf{x}_k^{(\text{new})} | \mathbf{x}_k, \mathbf{Z}_{k-1}) p(\mathbf{x}_k | \mathbf{Z}_{k-1}) d\mathbf{x}_k \quad (32)$$

If $\mathbf{x}_k$ and $\mathbf{x}_k^{(\text{new})}$ are independent, then Eq. (32) can be written in the form

$$\mathcal{PE} = \int p_{\text{new}}(\mathbf{x} | \mathbf{Z}_{k-1}) p_{\text{prior}}(\mathbf{x} | \mathbf{Z}_{k-1}) d\mathbf{x} \quad (33)$$

where $p_{\text{new}}(\cdot | \mathbf{Z}_{k-1})$ and $p_{\text{prior}}(\cdot | \mathbf{Z}_{k-1})$ are the PDFs (each conditioned on the prior reports $\mathbf{Z}_{k-1}$) of the new track and prior (predicted) tracks, respectively. If, in addition, $\mathbf{x}_k$ and $\mathbf{x}_k^{(\text{new})}$ are Gaussian with respective means $\boldsymbol{\mu}_1$ and $\boldsymbol{\mu}_2$ and respective covariances $\mathbf{P}_1$ and $\mathbf{P}_2$, then Eq. (33) reduces to

^{††}Alternatively, steps 3–5 of Algorithm 1 can be substituted with the solution of the full optimization problem (19).

$$-\ln \mathcal{PE} = \frac{1}{2}(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T (\mathbf{P}_1 + \mathbf{P}_2)^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) + \frac{1}{2} \ln \det(2\pi(\mathbf{P}_1 + \mathbf{P}_2)) \quad (34)$$

The first term in Eq. (34) involves the familiar Mahalanobis distance. However, for non-Gaussian uncertainties which commonly arise in space surveillance tracking due to the need to propagate over long time spans, higher fidelity filters and representations of the uncertainty must be used to account for possible higher-order statistics (cumulants). In other words, the prediction error is *filter dependent* and must be evaluated accordingly.

The association problem provides a natural application of the prediction error $\mathcal{PE}$ in Eq. (33) as a metric for uncertainty consistency. Referring to Fig. 5, suppose there are two satellite states at times t_1 and t_2 (with $t_1 \leq t_2$ without loss of generality) represented by the PDFs $p_1(\mathbf{x}, t_1)$ and $p_2(\mathbf{x}, t_2)$, respectively, and one wishes to assess whether the two objects associate. The likelihood score for computing the probability of association depends on the prediction error (33), which for this scenario is given by

$$\mathcal{PE}(t) = \int p_1(\mathbf{x}, t) p_2(\mathbf{x}, t) d\mathbf{x} \quad (35)$$

It is also convenient to define the “cost” metric from Eq. (35) according to

$$c(t) = -\ln \mathcal{PE}(t) \quad (36)$$

Evaluation of the integral Eq. (35) requires representations of both PDFs p_1 and p_2 at some common time t . Thus, one could either 1) propagate p_1 to time t_2 and evaluate $\mathcal{PE}(t_2)$, or 2) propagate p_1 from time t_1 to some intermediate time t_* , then back propagate p_2 to t_* , and finally evaluate $\mathcal{PE}(t_*)$. In fact, the value of the prediction error is *independent of the choice of t_** under the assumption that the dynamics are governed by a conservative deterministic model and the PDFs are propagated consistently. This result is formally stated and proved below.

Proposition 1. Let $p_1(\mathbf{x}, t)$ and $p_2(\mathbf{x}, t)$ denote the PDFs of two independent states at time t . Suppose the state $\mathbf{x}$ is governed by the conservative dynamical model

$$\mathbf{x}'(t) = \mathbf{f}(\mathbf{x}(t), t), \quad \nabla_{\mathbf{x}}^T \mathbf{f} = 0$$

and the time evolution of each PDF satisfies the noiseless Fokker–Planck–Kolmogorov equation

$$\frac{\partial p}{\partial t} = -\nabla_{\mathbf{x}}^T (p \mathbf{f}) \quad (37)$$

where $\nabla_{\mathbf{x}}$ is the gradient with respect to $\mathbf{x}$ viewed as a column operator. Then, the prediction error (35) is time-independent.

Proof: If p_1 and p_2 both satisfy Eq. (37), then the same can be said for their product. Indeed, by the product rule

$$\begin{aligned} \frac{\partial}{\partial t} (p_1 p_2) &= p_1 \frac{\partial p_2}{\partial t} + p_2 \frac{\partial p_1}{\partial t} = -p_1 \nabla_{\mathbf{x}}^T (p_2 \mathbf{f}) - p_2 \nabla_{\mathbf{x}}^T (p_1 \mathbf{f}) \\ &= -\nabla_{\mathbf{x}}^T (p_1 p_2 \mathbf{f}) \end{aligned}$$

Therefore

$$\begin{aligned} \frac{d}{dt} \mathcal{PE} &= \frac{d}{dt} \int p_1 p_2 d\mathbf{x} = \int \frac{\partial}{\partial t} (p_1 p_2) d\mathbf{x} \\ &= - \int \nabla_{\mathbf{x}}^T (p_1 p_2 \mathbf{f}) d\mathbf{x} = 0 \end{aligned}$$

as follows from the Gauss divergence theorem from elementary vector calculus. $\square$

The metric (35) provides a necessary condition for uncertainty consistency. If the evolution of the prediction error is not constant over time, then the propagated uncertainties of the two states are not consistent. The use of this metric for quantifying uncertainty consistency over long time gaps is illustrated in the next section.

In closing, the proof of proposition 1 can be replicated for other “distance metrics” such as the Kullback–Leibler divergence (KLD),

$$\mathcal{KL}(t) = \int p_1(\mathbf{x}, t) \ln \left[\frac{p_1(\mathbf{x}, t)}{p_2(\mathbf{x}, t)} \right] d\mathbf{x}$$

The KLD can thus serve as an alternative metric for uncertainty consistency. While the KLD is used in sensor resource management as a measure of mutual information gain [20], it is not what is commonly used in tracking for scoring likelihoods of association. A rigorous derivation of the likelihood ratio used to score the association of a track or measurement to an orbit is provided, for example, in Poore [19]; it is precisely the prediction error $p(\mathbf{z}_k | \mathbf{Z}_{k-1})$ or Eq. (32) which appears in the ensuing expression and not the KLD.

Simulation Studies

In this section, results demonstrating the proposed GSF are presented. For a low-Earth-orbit (LEO) scenario, it is first shown how the standard UKF does not necessarily maintain uncertainty consistency during long-term propagation and how higher fidelity GSFs can improve this situation. Next, the impact of correct uncertainty management on the problems of association and anomaly detection is illustrated. Finally, a weight adaptation scheme for uncertainty propagation (full details of which are explained in the appendix) is applied to the GSF and is shown not to provide any additional accuracy of the forecast PDF.

The simulation scenario considers a canonical test problem involving two closely spaced objects (CSOs) in LEO whose initial uncertainties are Gaussian with respect to equinoctial orbital elements. The initial states of the objects at time $t = 0$ are:

$$\begin{aligned} a_1 &= 6980 \text{ km}, & h_1 &= k_1 = p_1 = q_1 = \ell_1 = 0 & (\text{Object 1}) \\ a_2 &= 7020 \text{ km}, & h_2 &= k_2 = p_2 = q_2 = \ell_2 = 0 & (\text{Object 2}) \end{aligned}$$

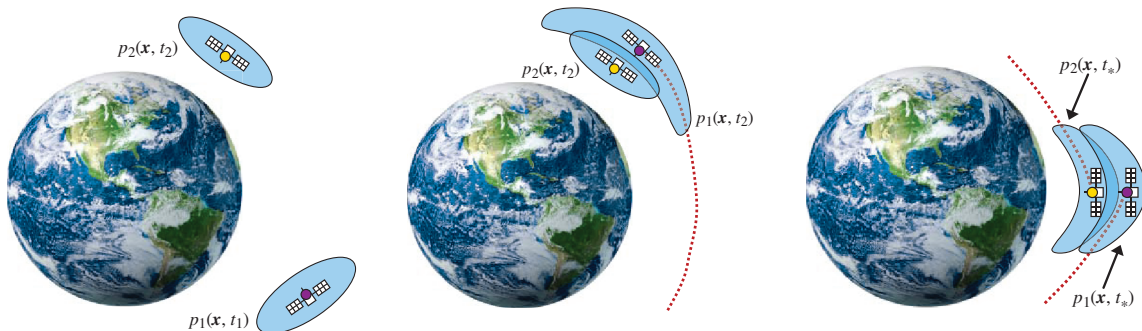


Fig. 5 The evaluation of the prediction error (35) used to compute the probability of association between two states at times t_1 and t_2 (left) requires representations of their PDFs at a common time t_* [e.g., $t_* = t_2$ (middle) or $t_1 < t_* < t_2$ (right)]. The prediction error is independent of the choice of t_* .

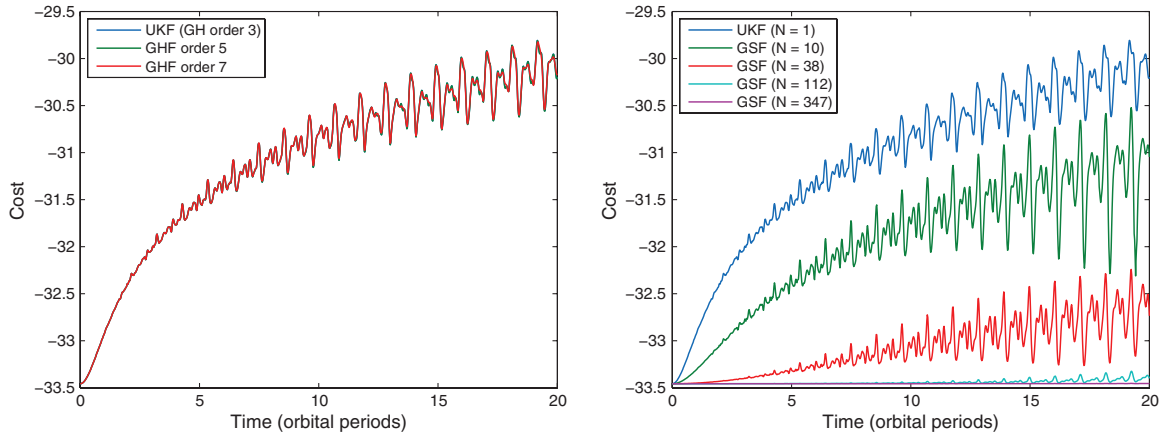


Fig. 6 Evolution of the cost metric (36) computed using the UKF, various GHFs, and various GSFs.

The initial covariances of the objects are $\mathbf{P}_1 = \mathbf{P}_2 = \mathbf{A}\mathbf{A}^T$, where^{††}

$$\mathbf{A} = \text{diag}\left(20 \text{ km}, 10^{-3}, 10^{-3}, 10^{-3}, 10^{-3}, 10^{-2} \frac{\pi}{180}\right)$$

From Eq. (35), it follows that the prediction error between the two objects at $t = 0$ is^{§§} $\mathcal{PE} = 3.387367558 \cdot 10^{14}$ with corresponding cost

$$-\ln \mathcal{PE} = -33.45624439 \quad (38)$$

Fixing a particular filter, the initial Gaussians are propagated^{¶¶} for a total time of 32.38 h (which is about 20 complete orbital periods) and the prediction error (35) and the cost (36) are evaluated at intermediate times. By proposition 1, the cost must retain the constant value given by Eq. (38) to assert that the state uncertainties are propagated consistently (i.e., a constant cost is a *necessary condition* for uncertainty consistency). Thus, any departure from a constant cost signals a degradation in the computed uncertainty.

The left half of Fig. 6 plots the evolution of the cost metric using the Gauss–Hermite filters (GHFs) of order three, five, and seven (the former is the standard UKF). Indeed, there is negligible difference in the computed costs. Therefore, the standard (third-order) UKF achieves *covariance consistency*. It must be emphasized that the GHFs compute the mean and covariance (of a possibly non-Gaussian PDF) directly from their respective definitions using Gauss–Hermite quadrature. By increasing the order the quadrature method (i.e., increasing the number of sigma points), the accuracy of the computed mean and covariance does not change. However, covariance consistency is only a prerequisite to the more general *uncertainty consistency*. Although the UKF correctly resolves the mean and covariance (first two cumulants) of the true PDF, it fails to capture information about the cumulants of third-order and higher. Because the cost diverges from the initial value, the true state uncertainties of the CSOs are evidently non-Gaussian and contain nonnegligible higher-order cumulants which impact the probability of association (POA). After 20 orbital periods, this divergence is nearly four units of cost or about a factor of 50 in the prediction error. Consequently, the resulting POA could have an error upwards of a full order of magnitude.

^{††}A value of 20 km for the uncertainty in the semimajor axis is representative of real data the authors have processed describing a breakup scenario in LEO. Later in this section, the rate of uncertainty degradation on the initial uncertainty in the semimajor axis is studied (see Fig. 7).

^{§§}The prediction error (or cost) depends on units. In this paper, the assumed length unit is Earth radii ($R_\oplus$); $1R_\oplus = 6378.137 \text{ km}$.

^{¶¶}Unless stated otherwise, the discrete-time square-root version of the UKF [15] is used to propagate Gaussian PDFs and component Gaussians within a Gaussian mixture. The UKF sigma points are propagated using the Astrodynamics Standards Special Perturbations (SP) program (Version 7) using a degree and order 70 gravity model.

The right half of Fig. 6 shows the improvement in uncertainty consistency when using the GSF with various numbers of mixture components. In generating these results, the initial Gaussian uncertainties of the CSOs were refined into Gaussian sums using Algorithm 1 and the component Gaussians propagated using the (third-order) UKF. For comparison purposes, the top curve is once again the cost obtained when the initial Gaussian uncertainties are propagated using the UKF. The remaining curves show the cost obtained using GSFs with 10, 38, 112, and 347 Gaussian components. By accounting for higher-order cumulants in the uncertainties, the GSF provides improved uncertainty consistency as seen by the mitigation of the divergence in the time evolution of the cost function. As the number of Gaussians increases, the uncertainty consistency improves. The highest fidelity filter ($N = 347$) shows no noticeable uncertainty degradation over the entire 20 orbital period propagation.

While the GSF can give high-accuracy results, it can be expensive as a potentially large number of Gaussians must be propagated. However, in some situations (e.g., short propagation times or small initial uncertainties in the semimajor axis a), such a fine level of accuracy may not be required. For example, Fig. 7 shows how the rate of uncertainty degradation for the UKF depends on the magnitude of the initial uncertainty (standard deviation) in the semimajor axis of the two CSOs. The dark blue (top) curve corresponds to an initial standard deviation in the semimajor axis of 20 km, which is the same value used to generate the plots in Fig. 6. The remaining curves are generated using a semimajor axis standard deviation of $A_{11} = \sigma_a$, where σ_a varies as shown in the figure. The initial means of the semimajor axes of the two CSOs are set^{***} to $a_1 = 7000 \text{ km} - \sigma_a$ and $a_2 = 7000 \text{ km} + \sigma_a$. Because the initial states and uncertainties differ, the prediction error or cost at time zero, as determined from Eq. (35), also differs for each σ_a case. Of greater interest is how these costs evolve in time. While the uncertainty consistency rapidly degrades for larger σ_a , with a small uncertainty of $\sigma_a = 1 \text{ km}$, uncertainty consistency is maintained for approximately 20 orbits. This result is consistent with Sabol et al. [17]. However, even for this case of small initial uncertainty in the semimajor axis, uncertainty consistency cannot be maintained indefinitely; within 50 orbital periods the degradation is significant. This indicates that for some situations (e.g., small σ_a , short propagation times) computationally cheap low-fidelity methods can be used, while more accurate methods like the Gaussian sum filter will be needed in other cases.

Figure 8 provides insight into the source of the periodic oscillations in the prediction error (cost). Here evolution of the cost metric is plotted using the UKF and various GSFs assuming propagation under either a full (SP) gravity model (thin curves) or unperturbed Keplerian dynamics (thick curves). The former set of curves are those from Fig. 6. For the Keplerian case, uncertainty

^{***}This choice of initial conditions ensures that the objects always start with a separation of $2\sigma_a$, though qualitatively the results are not dependent on this spacing.

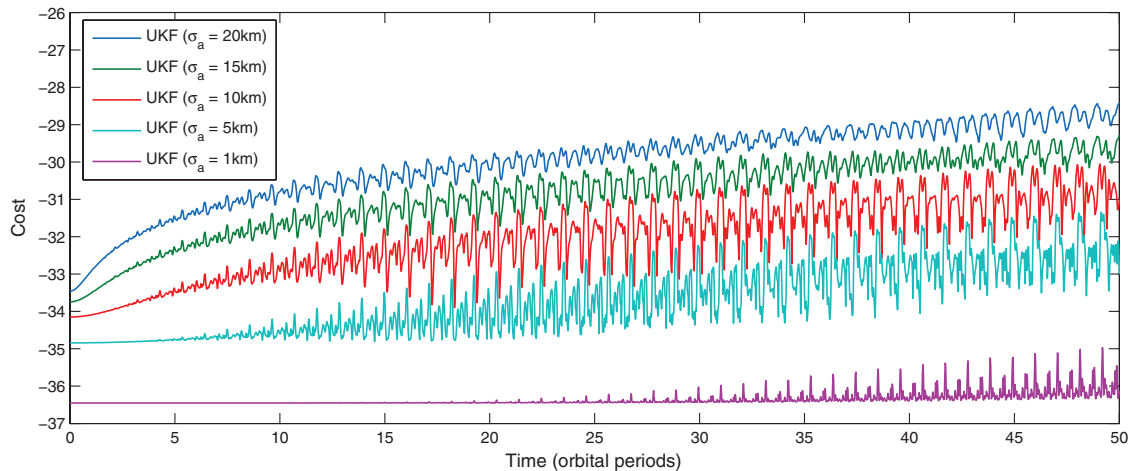


Fig. 7 Evolution of the cost metric (35) computed using the UKF comparing the effect of the size of the initial standard deviation σ_a in the semimajor axis of the two closely spaced objects.

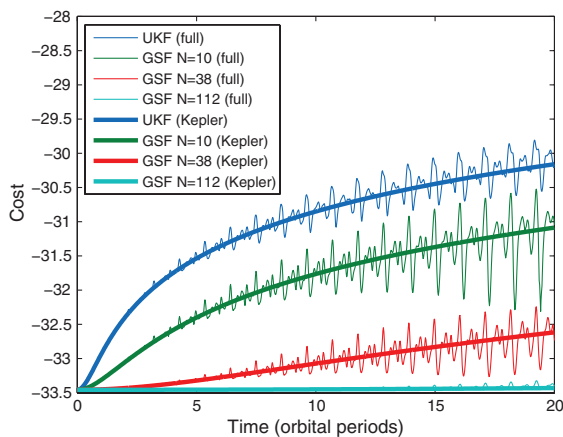


Fig. 8 Evolution of the cost metric (35) computed using the UKF and various GSFs where propagation is done assuming either a full gravity model (thin curves) or unperturbed Keplerian dynamics (thick curves).

consistency still degrades over time, but the degradation is essentially monotonic with a smoothly varying rate. The oscillations that appear when using the full gravity model are not present and it is conjectured that these *periodic* oscillations are due to the effects of the perturbations (and not a numerical stability issue). Thus, even under Keplerian dynamics, the initial Gaussian uncertainties evolve to something non-Gaussian (and are better described by considering higher-order representations of the uncertainty such as Gaussian sums).

Assuming use of the full gravity model for the remainder of this section, Fig. 9 illustrates the impact of uncertainty consistency when state uncertainties are represented in Cartesian ECI coordinates versus equinoctial orbital elements and alternate equinoctial orbital elements.^{†††} Here we can see that the cost metric degrades much more rapidly in Cartesian space; while there is an error of about 5.5 units of cost over 50 orbital periods in equinoctial orbital elements, this same error is surpassed in less than a single orbital period in ECI. It is also important to note that both the ECI and orbital element implementations of the UKF used here assume the square-root version of the UKF [15]. This greatly improves numerical conditioning and stability, allowing a Gaussian to be propagated over a long time period in ECI (and orbital elements) without suffering from “covariance collapse.” Thus, in defense of ECI, the standard UKF

still maintains covariance consistency (since there is little change when using higher-order Gauss–Hermite filters).

Figures 9 and 10 also show the additional benefit of representing and propagating uncertainties in the alternate set of equinoctial orbital elements over the traditional set. For both the traditional and alternate element sets, the use of Gaussian sums mitigates the degradation in the evolved prediction error (cost) and hence improves the consistency (accuracy) of the represented state uncertainty. For long propagation times, the degradation in cost is slightly less when using the alternate set over the traditional set. However, around $t = 0$, the cost curves in the alternate set are significantly flatter than those in the traditional set. Thus, for short propagation times, the use of alternate equinoctial orbital elements provides a more accurate representation of the state PDF compared with representations in traditional equinoctial orbital elements.

Impact on Association and Anomaly Detection

The example in Fig. 11 demonstrates the impact of correct uncertainty management on the problems of association (correlation) and anomaly (maneuver, change) detection. The left half of the figure depicts the non-Gaussianity of the uncertainty in Object 1 after only four orbital periods (about 6.5 h). A two-dimensional slice of its PDF along the semimajor axis a and mean longitude ℓ coordinates is plotted. The true uncertainty (grayscale colormap) was computed using a high-fidelity Gaussian sum filter with $N = 1127$ and is contrasted with the Gaussian uncertainty (reddish colormap) obtained from the UKF. Suppose one wishes to compute the prediction error (or cost) associated with a new Gaussian track state (blueish colormap) located between the tails of the true distribution. In this particular example, a cost of -32.675 is obtained from the UKF approximation versus a cost of $+3.5595$ from the $N = 1127$ GSF. Therefore, the UKF estimates a smaller cost leading to a higher probability of association and *possibly a misassociation* and *possibly a failure to detect an anomaly*. Note that even a medium fidelity GSF (e.g., 112 or 347 terms) can estimate the true uncertainty and hence the cost much more accurately than the UKF, and this in turn mitigates misassociations and undetected anomalies.

Effect of Weight Update Scheme

Figure 12 shows the evolution of the cost metric (36) using various GSFs and the impact of using a weight adaptation method for uncertainty propagation. Unperturbed Keplerian dynamics are assumed in this figure. The solid curves correspond to the GSF of this paper, which does not use any weight update method by default (these curves are the same as the thick curves in Fig. 8). The dashed curves in Fig. 6 show the evolved cost when using a GSF in conjunction with the weight update scheme described in the appendix with the updates occurring at a frequency of every one-tenth of an orbital period (i.e., 200 times over the length of the

^{†††}For this comparison, the initial uncertainty is assumed to be Gaussian in all coordinate systems. For the ECI and alternate equinoctial cases, the initial uncertainty was computed by performing an unscented transform on the initial Gaussian (in equinoctial elements) defined at the beginning of this section.

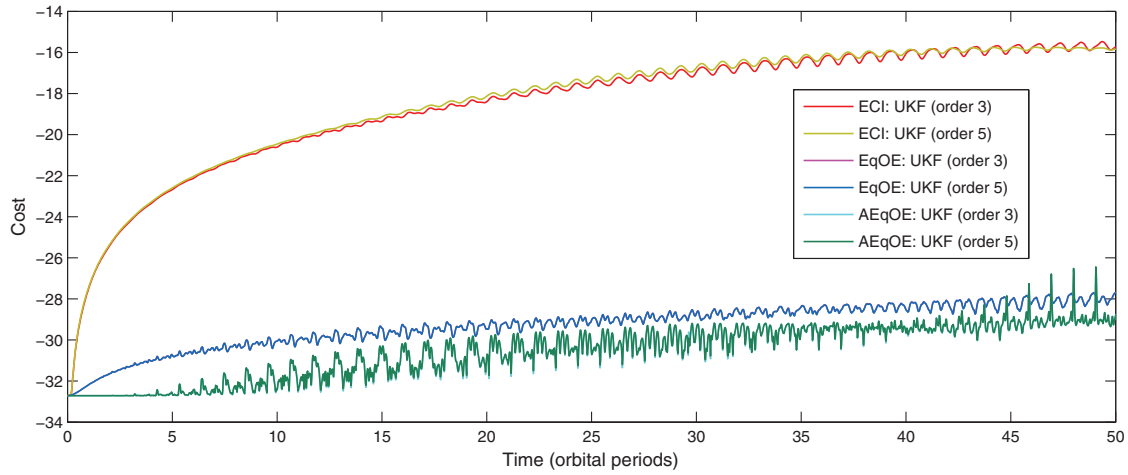


Fig. 9 Evolution of the cost metric (35) computed using the UKF where the uncertainty is represented in Cartesian ECI coordinates versus equinoctial orbital elements (EqOE) and alternate equinoctial orbital elements (AEqOE).

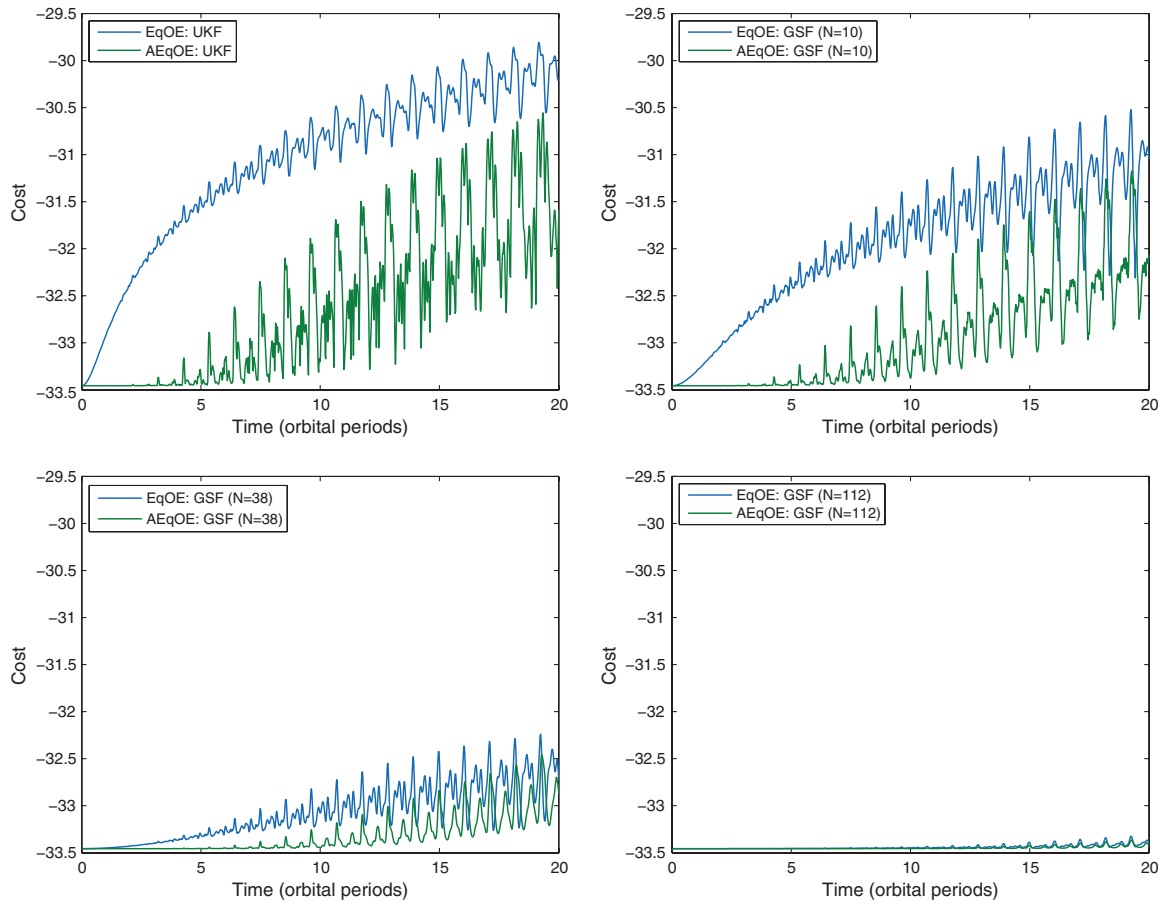
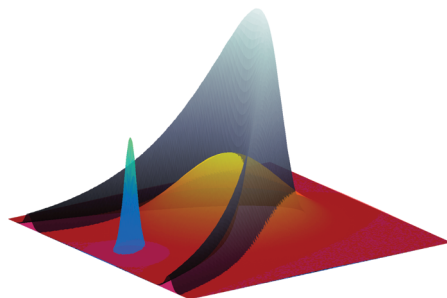


Fig. 10 Evolution of the cost metric (35) computed using the UKF and various GSFs where the uncertainty is represented in (traditional) EqOE versus AEqOE.



No. of Gaussians	$-\ln \mathcal{PE}$
1	-32.675
10	-22.835
38	-0.34305
112	3.4919
347	3.5592
1127	3.5595

Prediction error (cost) computed from the Gaussian sum filter

Fig. 11 (Left) The PDF of Object 1 in the semimajor axis a and mean longitude ℓ coordinates after four orbital periods computed using the UKF (reddish colormap) and a GSF with $N = 1127$ (grayscale colormap). The blueish Gaussian represents a track update. The prediction error for the update against various fidelity GSFs are shown in the table (right).

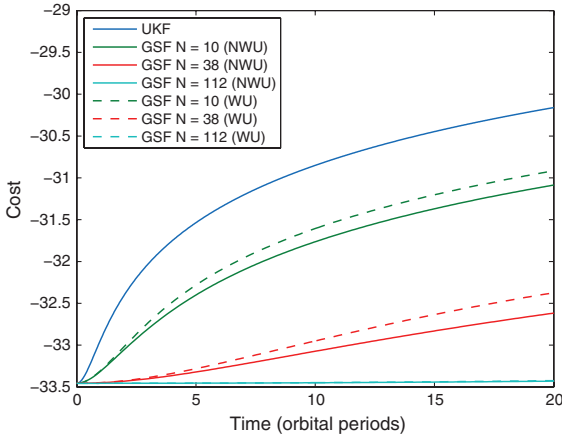


Fig. 12 Evolution of the cost metric (36) computed using the unscented Kalman filter and various GSFs, and applying either no weight update method or the weight update method outlined in the appendix.

scenario). Although the dashed curves (which use the weight update) agree at early times with the solid curves (which do not use any weight update), there is some divergence at later times, especially for the low ($N = 10$) and medium ($N = 38$) fidelity Gaussian sums. In all cases, the application of the weight adaptation scheme does not improve uncertainty consistency.

Conclusions

In this paper, an algorithm for choosing the component means, covariances, and weights of a Gaussian sum was proposed with the property that the standard UKF, when acting in parallel on each component, maintains uncertainty consistency up to a prescribed accuracy. The algorithm is based on the observation that the rate at which the uncertainty becomes non-Gaussian is dictated by the initial uncertainty in the semimajor axis (radial) direction. Superior computational efficiency is realized because the underlying optimization problem that determines the initial Gaussian components only has to be performed once *offline*. In particular, with the optimal solution of Eq. (19), one can either substantially reduce the error in the Gaussian sum approximation or, for the same error, reduce the number of Gaussians needed to achieve a prescribed accuracy. Consequently, the additional application of an *online* weight adaptation scheme for Gaussian sum uncertainty propagation did not provide any additional improvement in uncertainty consistency.

Equinoctial orbital elements are an efficient coordinate system for representing Gaussian sums because the refinement can essentially be done only along the radial (semimajor axis) coordinate. Such a computational advantage would not be realizable in ECI coordinates because refinement would be required along all six state-space coordinates and, as was observed, Gaussian PDFs become non-Gaussian more quickly when represented in ECI than when represented in equinoctial elements. An alternate set of equinoctial orbital elements, which provided superior accuracy and uncertainty consistency over the traditional set, especially over shorter propagation times, was also proposed. Finally, it was observed that a poor representation of a non-Gaussian state can lead to an inaccurate prediction error. This can adversely impact data association (correlation), sensor resource management, collision avoidance, and anomaly detection.

Appendix

This appendix summarizes the L^2 based “Weight Update I” adaptation method for Gaussian sum propagation proposed in Terejanu et al. [6]. In what follows, the method is specialized to the unperturbed two-body problem (with no process noise) in equinoctial orbital elements. This simple unperturbed problem provides a baseline for testing this scheme against the proposed Gaussian sum filter of this paper.

The method assumes discrete-time dynamics and forms new weights by minimizing the L^2 error between the exact solution governed by the Chapman–Kolmogorov equation and the Gaussian sum approximation. At the initial time $t = t_0$, it is assumed that the state PDF is Gaussian with respect to equinoctial orbital elements $\mathbf{x} = (a, h, k, p, q, \ell)$ with mean $\mathbf{v}_0$ and covariance $\mathbf{Q}_0$; i.e.

$$p_{\text{true}}(\mathbf{x}, t_0) = \mathcal{N}(\mathbf{x}; \mathbf{v}_0, \mathbf{Q}_0)$$

where $\mathcal{N}$ is the Gaussian PDF defined by Eq. (12). For deterministic dynamics, it follows that the exact PDF at any future time $t > t_0$ is given by [21]

$$p_{\text{true}}(\mathbf{x}, t) = \mathcal{N}(\boldsymbol{\psi}(\mathbf{x}, t; t_0); \mathbf{v}_0, \mathbf{Q}_0)$$

where $\boldsymbol{\psi}$ denotes the inverse solution flow from dynamical systems theory. For the unperturbed two-body problem in equinoctial elements, it follows that

$$\boldsymbol{\psi}(\mathbf{x}, t; t_0) = (a, h, k, p, q, \ell - n(t - t_0))^T \quad (\text{A1})$$

where $n = \sqrt{\mu_{\oplus}/a^3}$ is the mean motion and $\mu_{\oplus}$ is the Earth gravitational constant. Let

$$p_{\text{approx}}(\mathbf{x}, t) = \sum_{\alpha=1}^N w_{\alpha} \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_{\alpha}, \mathbf{P}_{\alpha})$$

be a Gaussian sum approximation to $p_{\text{true}}(\mathbf{x}, t)$. The updated weights $\tilde{w}_1, \dots, \tilde{w}_N$ are taken to be the solution of the following L^2 optimization problem:

$$\begin{aligned} \tilde{\mathbf{w}} \equiv (\tilde{w}_1, \dots, \tilde{w}_N)^T = & \text{Minimize}_{w_1, \dots, w_N} \frac{1}{2} \int [p_{\text{true}}(\mathbf{x}, t) \\ & - p_{\text{approx}}(\mathbf{x}, t)]^2 d\mathbf{x} \\ \sum_{\alpha=1}^N w_{\alpha} = & 1, \quad \text{Subject to } w_{\alpha} \geq 0, \quad \alpha = 1, \dots, N \end{aligned} \quad (\text{A2})$$

Upon noting the identity (20), the optimization problem (A2) can be expressed as a quadratic programming problem (QPP) in the form

$$\begin{aligned} \tilde{\mathbf{w}} = & \text{Minimize}_{w_1, \dots, w_N} \frac{1}{2} \mathbf{w}^T \mathbf{M} \mathbf{w} - \mathbf{w}^T \mathbf{n} \\ \text{Subject to } & \sum_{\alpha=1}^N w_{\alpha} = 1, \quad w_{\alpha} \geq 0, \quad \alpha = 1, \dots, N \end{aligned} \quad (\text{A3})$$

where the matrix $\mathbf{M} \in \mathbb{R}^{N \times N}$ and vector $\mathbf{n} \in \mathbb{R}^N$ have components

$$(\mathbf{M})_{\alpha\beta} = \mathcal{N}(\boldsymbol{\mu}_{\alpha} - \boldsymbol{\mu}_{\beta}; \mathbf{0}, \mathbf{P}_{\alpha} + \mathbf{P}_{\beta}) \quad (\text{A4})$$

$$(\mathbf{n})_{\alpha} = \int \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_{\alpha}, \mathbf{P}_{\alpha}) \mathcal{N}(\boldsymbol{\psi}(\mathbf{x}, t; t_0); \mathbf{v}_0, \mathbf{Q}_0) d\mathbf{x} \quad (\text{A5})$$

The computation of the integral (A5) is the only nontrivial task in setting up the QPP (A3). For $\boldsymbol{\psi}$ given by Eq. (A1), the logarithm of the integrand in Eq. (A5) is quadratic in the last five equinoctial elements (h, k, p, q, ℓ). Therefore, integration in these coordinates can be done analytically reducing Eq. (A5) to

$$(\mathbf{n})_{\alpha} = c_{\alpha} \int_{-\infty}^{\infty} e^{-g_{\alpha}(a)} da \quad (\text{A6})$$

for some constant c_{α} and some function g_{α} dependent on the semimajor axis a . This resulting one-dimensional integral can be evaluated using Gauss–Hermite quadrature. In doing so, it is necessary to choose the correct Gaussian weight function in order that the quadrature nodes sample the integrand over an appropriate region. Following Liu and Pierce [22], Eq. (A6) can be rewritten as

$$(\mathbf{n})_{\alpha} = \int \mathcal{N}(a; a_*, \sigma_*^2) h_{\alpha}(a) da \quad (\text{A7})$$

where

$$h_{\alpha}(a) = c_{\alpha} e^{-g_{\alpha}(a)} / \mathcal{N}(a; a_*, \sigma_*^2)$$

and $g'_{\alpha}(a_*) = 0$ and $\sigma_*^2 = 1/g''_{\alpha}(a_*)$. For the test case considered in the simulations section, the functions $h_{\alpha}(a)$ are not well represented by a low-order polynomial. To achieve five decimal digits of accuracy in the components of $\mathbf{n}$, a 99-point Gauss–Hermite quadrature method was required.

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